

Mohamad Latifian

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Research Interests

My research lies at the intersection of theoretical computer science, artificial intelligence, and economics, with a focus on algorithms and computational complexity. My primary interests include computational social choice, fair division, algorithmic game theory, and mechanism design.

Education and Academic Positions

Postdoctoral Research Associate

University of Edinburgh, UK

Host: Aris Filos-Ratsikas

Jul. 2024 – present

Ph.D. in Computer Science

University of Toronto, Canada

Advisors: Nisarg Shah and Allan Borodin

Thesis: Decoding Distortion: Pushing the Limits of Optimal Collective Decision-Making

GPA: 4.0/4.0

Sep. 2020 – Sep. 2024

M.Sc. in Algorithms and Computation

Sharif University of Technology, Iran

Advisor: Mohammad Ghodsi

Thesis: Considering the Effect of Abstention in the Distortion Value.

GPA: 19.59/20

Sep. 2017 – Sep. 2019

B.Sc. in Computer Engineering

Sharif University of Technology, Iran

Advisor: Hamid Zarrabi-Zadeh

Thesis: Hamiltonian Path in Solid Grid Graphs.

GPA: 17.22/20

Sep. 2012 – Jul. 2017

Conference Publications

(Authors in alphabetical order)

- C19. *Nash Equilibria in Auctions with Pacing Strategies: Complexity and Inefficiency*. A. Filos-Ratsikas, C. Kokkalis, and M. Latifian.
Proc. of the 22nd Conference on Web and Internet Economics (**WINE**), 2026, (Forthcoming).
- C18. *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions*. D. Afazeli, M. Latifian, and M. Seddighin.
Proc. of the ACM Web Conference 2026 (**WWW**), pp. 375–383, 2026.
- C17. *Metric Distortion with Preference Intensities*. M. Abbaszadeh, A. Ansarifard, M. Latifian, and M. Seddighin.
Proc. of the 40th AAAI Conference on Artificial Intelligence (**AAAI**), volume 40, pp. 16603–16611, 2026.
- C16. *Optimal Metric Distortion for Matching on the Line*. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris.
Proc. of the 34th International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 3857–3865, 2025.
- C15. *Towards Fair and Efficient Public Transportation: A Bus Stop Model*. M. Bullinger, E. Elkind, and M. Latifian.
Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 427–435, 2025.

- C14. *Temporal Fair Division of Indivisible Items*. E. Elkind, A. Lam, M. Latifian, T. Y. Neoh, and N. Teh.
Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 676–685, 2025.
- C13. *Fair Division with Market Values*. S. Barman, S. Ebadian, M. Latifian, and N. Shah.
Proc. of the 39th AAAI Conference on Artificial Intelligence (AAAI), volume 39, pp. 13589–13596, 2025.
- C12. *The Distortion of Public-Spirited Participatory Budgeting*. M. Bedaywi, B. Flanigan, M. Latifian, and N. Shah.
Proc. of the 39th AAAI Conference on Artificial Intelligence (AAAI), volume 39, pp. 13605–13613, 2025.
- C11. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris.
Proc. of the 33rd International Joint Conference on Artificial Intelligence (IJCAI), pp. 2851–2859, 2024.
- C10. *Computational Aspects of Distortion*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah.
Proc. of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 499–512, 2024.
- C9. *Explainable and Efficient Randomized Voting Rules*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah.
Proc. of the 37th Conference on Neural Information Processing Systems (NeurIPS), volume 36, 2023.
- C8. *Best of Both Distortion Worlds*. V. Gkatzelis, M. Latifian, and N. Shah.
Proc. of the 24th ACM Conference on Economics and Computation (EC), pp. 738–758, 2023.
- C7. *The Distortion of Approval Voting with Runoff*. S. Ebadian, M. Latifian, and N. Shah.
Proc. of the 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 1752–1760, 2023.
- C6. *Voting with Preference Intensities*. A. Kahng, M. Latifian, and N. Shah.
Proc. of the 37th AAAI Conference on Artificial Intelligence (AAAI), pp. 5697–5704, 2023.
- C5. *Distortion in Voting with Top-t Preferences*. A. Borodin, D. Halpern, M. Latifian, and N. Shah.
Proc. of the 31st International Joint Conference on Artificial Intelligence (IJCAI), pp. 116–122, 2022.
- C4. *Almost Envy-freeness, Envy-rank, and Nash Social Welfare Matchings*. A. Farhadi, M. Hajiaghayi, M. Latifian, M. Seddighin, and H. Yami.
Proc. of the 35th AAAI Conference on Artificial Intelligence (AAAI), pp. 5355–5362, 2021.
- C3. *Resource-Aware Protocols for Network Cost-Sharing Games*. G. Christodoulou, V. Gkatzelis, M. Latifian, and A. Sgouritsa.
Proc. of the 21st ACM Conference on Economics and Computation (EC), pp. 81–107, 2020.
- C2. *On the Distortion Value of the Elections with Abstention*. M. Ghodsi, M. Latifian, and M. Seddighin.
Proc. of the 33rd AAAI Conference on Artificial Intelligence (AAAI), pp. 1981–1988, 2019.
- C1. *Rent Division Among Groups*. M. Ghodsi, M. Latifian, A. Mohammadi, S. Moradian, and M. Seddighin.
Proc. of the 12th International Conference on Combinatorial Optimization and Applications (COCOA), pp. 577–591, 2018.

Journal Publications

- J4. *Optimal Metric Distortion for Matching on the Line*. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris. *Artificial Intelligence*, (Forthcoming).
- J3. *Computational Aspects of Distortion*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. *Autonomous Agents and Multi-Agent Systems*, (Forthcoming).
- J2. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris. *Autonomous Agents and Multi-Agent Systems*, 39(2):42, 2025.
- J1. *On the Distortion Value of Elections with Abstention*. M. Ghodsi, M. Latifian, and M. Seddighin. *Journal of Artificial Intelligence Research*, 70:567–595, 2021.

Working Papers

(Authors in alphabetical order)

- M5. *Group Assignment: Bridging Voting and Matching through Distortion*.
M. Latifian.
- M4. *Capturing the Effect of Randomization in Distortion*.
M. Latifian.
- M3. *Envy-free Cake Cutting for Five Agents*.
A. Filos-Ratsikas, A. Hollender, and M. Latifian.
- M2. *Metric Distortion with Graph Objectives*.
A. Filos-Ratsikas, M. H. Ghasemi, M. Latifian, and M. Sharify.
- M1. *Distortion of Truthful Matching Mechanisms*.
V. Gkatzelis, M. Latifian, and E. Rewinski.

Academic Visits

Long-Term

University of Oxford, UK.
Host: Edith Elkind

Sep. – Nov. 2023

Max Planck Institute for Informatics, Germany.
Host: Alkmini Sgouritsa

May – Jun. 2019

Work title: *Resource-Aware Protocols for Network Cost-Sharing Games* ([C3.](#))

Short-Term

University of Oxford, UK.
Host: Alexandros Hollender

Sep. 2025

University of Edinburgh, UK.
Host: Aris Filos-Ratsikas

Nov. 2023

University of Essex, UK.
Host: Alexandros Voudouris

Oct. 2023

Professional Service

- **PC Member:** AAAI '27, EC '26, AAAI '26, EC '25, AAMAS '25, ICLR '25, AAAI '25, NeurIPS '24, IJCAI '24, AAMAS '24, AAAI '24, AAAI '21
- **Journal Reviewer:** Autonomous Agents and Multi-Agent Systems (JAAMAS), Artificial Intelligence (AIJ), Journal of Artificial Intelligence Research (JAIR)
- **Conference Reviewer:** SODA '27, ESA '26, STOC '25

Invited Talks

- T7. *Fair Division with Market Values*, Workshop on Fair Resource Allocation: Concepts, Algorithms and Complexity (FaiRACAC '25), held in conjunction with ICALP 2025 (7 Jul. 2025).
- T6. *Distortion in Social Choice Problems*, University of Edinburgh, EconCS Group Meetings (Oct. 2024).
- T5. *Fair Division with Market Values*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Lab Lunch (Oct. 2024).
- T4. *How to Make Efficient (Collective) Decisions*, Tehran Institute for Advanced Studies (Sep. 2024).
- T3. *Best of Both Distortion Worlds*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Seminar (Nov. 2023).
- T2. *Best of Both Distortion Worlds*, University of Essex, The Centre for Computational Finance and Economic Agents (CCFEA) Seminar Series (Oct. 2023)
- T1. *Resource-Aware Protocols for Network Cost-Sharing Games*, The Drexel EconCS Research Group Meetings (Jun. 2020).

Selected Teaching Experience

Guest Lecturer Oct. 2025
 University of Edinburgh
 Algorithms and Data Structures ([Course web page](#))

Teaching Assistant
 University of Toronto

- **Algorithm Design, Analysis & Complexity** Fall '23
- **Data Structures and Analysis** Winter '23
- **Algorithmic Game Theory and Mechanism Design** Fall '22

Teaching Assistant (* := Lead TA)
 Sharif University of Technology

- **Algorithmic Game Theory** Fall '17, Spring '19*
- **Design and Analysis of Algorithms** Fall '14, Fall '18*
- **Theory of Languages and Automata** Fall '14, Spring '15, Fall '15, Spring '16
- **Data Structures and Algorithms** Spring '15

Member of the Scientific Committee Fall 2016
 4th Sharif National Students Competition,
 Sharif University of Technology.

Head of the Scientific Committee Aug. 2013 and Aug. 2014
 13th & 14th Mehregan Mathematics Workshop,
 Shahid Ejeii High School.

Teaching Special Topics in Mathematics

2011–2013

*Teaching high-school students preparing for Olympiad in Informatics.**Topics included Algorithms, Graph Theory, Problem-Solving Strategies, and Programming.***Supervision Experience**

Mohammad Hossein Ghasemi and Mohammad Sharify (Co-supervised with A. Filos-Ratsikas)

Undergraduate students, Sharif University of Technology

Aug. 2025 – present

Title: *Metric Distortion with Graph Objectives*

Hamidreza Alipour, Kiarash Joolaei, Sepehr Mizanian

Undergraduate students, Sharif University of Technology

Nov. 2024 – present

Title: *A Deeper Look at Distortion in Impartial Culture Electorates*

Dana Afazeli

Undergraduate student, Sharif University of Technology

Aug. 2024 – Jan. 2026

Title: *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions (Appeared in WWW '26)*

Mehrad Abbaszadeh and Ali Ansarifar (Co-supervised with M. Seddighin)

Undergraduate students, Sharif University of Technology

Oct. 2023 – Aug. 2025

Title: *Metric Distortion with Preference Intensities (Appeared in AAAI '26)***Honors and Awards**

- **C.C. Gotlieb (Kelly) Graduate Fellowship** in the Department of Computer Science, University of Toronto 2023
- **Silver medal** in the 21st Iranian National Olympiad in Informatics (INOI). 2011
- **Ranked 2nd and 7th** in Nationwide Graduate Entrance Exam, in Algorithms and Computation and Artificial Intelligence majors, respectively. 2017
- **Ranked 1st** in cumulative GPA among 97 M.Sc. students of the department, Sharif University of Technology. 2019
- Scholarship from Iranian National Elites Foundation 2012 – 2019

Activities

- **Elected Member & President** of the Students' Scientific Chapter (SSC) Computer Engineering Department of Sharif University of Technology. 2014 – 2015
- **Chief of Staff** of 17th & 18th International Collegiate Programming Contest (ICPC) Asia Region - Tehran Site. 2015 – 2016
- **Chief of Staff** of 6th & 7th AI Challenge An event where AI agents written in Java, C++, or Python compete to pass the game field and overcome other agents. 2014 – 2015